tools.

What becomes most apparent when doing this exercise is that the React version of both pages is significantly more verbose than the Hotwire version. It's a little hard to compare lines of code between JavaScript, TypeScript, Ruby, JSX, and ERB. But the React versions of both pages just contain way more lines of code. The Hotwire version of the venue page was about 300 lines of code, replacing about 500 lines of React/JSX. When you consider that a large chunk of that 300 is just the same markup translated, that's a very big reduction in code logic. (It's probably also worth mentioning that my previous Stimulus-without-Hotwire take on this page was about 400 lines of code.) If anything, the difference on the schedule page is even larger; there's a lot of React code needed on that page to cover the things that Hotwire did in a couple of lines.

When looking at the difference in code, we see it comes from a few different sources:

- *TypeScript boilerplate.* React encourages the creation of more domain objects, and in conjunction with TypeScript, that means more code devoted to defining types. This is especially true given the way we wrote interfaces for each individual action object in the reducers.

- *React boilerplate.* This includes component definitions, calling hooks, managing the reducers, plus a little extra server-side code to shape the data. It's a general feature of JavaScript style that there is more typed boilerplate than in Rails style. Rails tends to avoid boilerplate code where possible; most JavaScript frameworks tend to have more of it. Also, React encourages smaller components, which means more boilerplate in terms of declarations, headers, and the like.

- *Extra state.* The React code requires us to keep a whole bunch of system state client side, nearly all of which is arguably duplicative of either the server state, or the state of the DOM itself. Another way of saying that last part is that in React, the DOM reflects the state of the client, whereas in Hotwire, the DOM is itself the state of the client.